The Effect of Background Music on Perceived Productivity Among High School Students

Doral Academy

AP Statistics

Ms. Zogovic

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Abstract

This study investigated whether listening to music while studying affects students' perceived productivity. A sample of 66 upperclassmen from Doral Academy was selected using a multistage sampling method and surveyed using a Google Form questionnaire. Students rated their perceived productivity on a 1 to 5 Likert scale. Of the participants, 48 reported studying with music and 18 reported studying in silence. A two-sample t-test was conducted to compare the mean productivity scores between the two groups. Students who studied with music reported a mean productivity score of 4.125, while students who studied in silence reported a mean score of 2.611. The observed difference of 1.514 points was statistically significant, $t(21) = 4.40$, $p < .001$. A 95% confidence interval for the difference in means was (0.799, 2.229). These findings provide statistical evidence of an association between listening to music while studying and increased perceived productivity among high school students.

Introduction and Importance

Students often search for strategies that may improve concentration, increase efficiency, and reduce stress while studying. One commonly used strategy is listening to background music. Some students believe music improves focus and motivation, while others believe silence is the most effective study environment. As streaming platforms and headphones become increasingly accessible, music has become a regular part of many students' study routines.

Understanding whether music affects perceived productivity is important for both students and educators. If music positively influences how productive students feel while studying, it may serve as a useful motivational tool. However, if music distracts students, it could negatively affect learning and academic performance. Because study habits directly impact academic success, investigating the relationship between music and productivity has educational relevance.

Previous research suggests that background music may improve concentration and academic performance for some students, particularly when the music is instrumental or minimally distracting. Kumar et al. (2016) found that many students reported improved focus while listening to music, while Muslimah and Apriani (2020) reported that over 80% of participants believed music improved concentration. These findings support further investigation into how music affects students' perceived productivity.

Research Question and Hypotheses

The research question for this study was:

Does listening to music while studying affect students' perceived productivity compared to studying in silence?

The hypotheses were:

H_0: $\mu music = \mu silence$

There is no difference in mean perceived productivity between students who study with music and students who study in silence.

H_a: $\mu music \neq \mu silence$

There is a difference in mean perceived productivity between students who study with music and students who study in silence.

Variables

The independent variable in this study was the study environment, categorized as studying with music or studying in silence. Students self-identified which environment they typically used while studying.

The dependent variable was perceived productivity, measured using a five-point Likert scale in which 1 represented very low productivity and 5 represented very high productivity.

Additional survey questions collected contextual information regarding grade level, study duration, and music preferences.

Methodology

This study used a quantitative survey design to examine whether listening to music while studying affects students' perceived productivity. The target population consisted of high school students at Doral Academy, while the sampling frame included juniors and seniors enrolled in selected classes.

A multistage sampling method was used. First, one building containing approximately 40 teachers was randomly selected from the school. Four teachers from that building were then randomly chosen using a random number generator. Systematic sampling was applied to the class rosters by selecting every ninth student after a random starting point until a total of 66 students had been selected.

Participants completed an anonymous Google Form survey independently using a QR code. Participation was voluntary, and no identifying information was collected. This helped minimize risks to participants and maintain confidentiality.

The survey measured perceived productivity using a 1 to 5 Likert scale. Additional questions gathered information regarding music type, listening method, and study habits.

Potential Sources of Bias

Several potential sources of bias should be considered. Because participants came from selected classes within one school, the sample may not fully represent all students at Doral Academy. Additionally, the study relied on self-reported measures of productivity, which may not accurately reflect actual academic performance. Students may also have overstated their productivity due to social desirability bias. Finally, several confounding variables such as academic ability, personality, study environment, and course difficulty were not controlled.

Results and Statistical Analysis

A total of 66 students participated in the study. Of those students, 48 reported studying with music and 18 reported studying in silence.

Students who studied with music reported a mean productivity score of 4.125 with a standard deviation of 0.789. Students who studied in silence reported a mean productivity score of 2.611 with a standard deviation of 1.378.

Study Environment	Sample Size	Mean Productivity	Standard Deviation
Music	48	4.125	0.789
Silence	18	2.611	1.378

A Welch two-sample t-test was conducted to compare the mean productivity scores between the two groups.

The conditions for inference were reasonably satisfied. Individual responses were independent, the sample size was sufficiently large, and Doral Academy's student population exceeded the 10% condition requirement.

The observed difference in means was:

$4.125 - 2.611 = 1.514$

The standard error was calculated to be 0.344, producing a test statistic of:

$t = 4.40$

with approximately 21 degrees of freedom.

The p-value was approximately 0.00024.

Because the p-value was less than 0.05, the null hypothesis was rejected. There is strong statistical evidence that students who study with music report different levels of perceived productivity compared to students who study in silence.

A 95% confidence interval for the difference in means was calculated as:

(0.799, 2.229)

Because the entire interval is positive and does not include zero, the confidence interval supports the conclusion that students who study with music report higher perceived productivity on average.

Discussion

The results of this study indicate that students who study with music report substantially higher levels of perceived productivity than students who study in silence. The difference between the two groups was both statistically significant and practically meaningful, with students who listened to music averaging approximately 1.5 points higher on a five-point productivity scale.

These findings are consistent with previous research suggesting that moderate background stimulation may improve focus and motivation for certain learners. Music may help reduce boredom, increase alertness, and create a more comfortable study environment. However, because this study measured self-perceived productivity rather than actual academic performance, the findings are more closely related to students' perceptions and motivation than direct measures of achievement.

Despite the strong statistical evidence, several limitations should be considered. The sample was limited to students from selected classes within one school, which restricts generalizability. Additionally, productivity was measured using self-reported survey responses, which may be influenced by inaccurate self-assessment or social desirability bias. Confounding variables such as personality traits, academic ability, and study environment were also not controlled.

Because this study was observational and not experimental, causation cannot be established. The study demonstrates an association between listening to music and perceived productivity, but it does not prove that music directly causes increased productivity.

Future research could use experimental designs in which students are randomly assigned to music or silence conditions while completing standardized academic tasks. Additional studies could also compare different music genres and examine objective academic outcomes such as quiz scores, accuracy, or study duration.

Conclusion

The purpose of this study was to determine whether listening to music while studying affects students' perceived productivity. Students who reported studying with music had significantly higher productivity ratings than students who studied in silence. Statistical analysis showed a significant difference between the two groups, with music listeners averaging approximately 1.5 points higher on a five-point productivity scale.

Although the study cannot establish causation, the results suggest that background music may positively influence how productive students feel while studying. These findings contribute to the ongoing discussion regarding effective study environments and individualized learning strategies.

References

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Raw survey data is available at:

https://github.com/supad002-ux/AP-Statistics-ASA-Project/blob/main/AP_Statistics_ASA_Project1_Music_and_Productivity